

Efficiency of carbonized *Sapindus emarginatus* as a low cost biodegradable adsorbent for the removal of safranine and methylene blue from aqueous solution

P. Mercy Jasper¹, S. Sumithra² and M.Madakka^{3*}

^{1*} & ²Department of Environmental Sciences, Yogi Vemana University, Kadapa, Andhra Pradesh, India.

³Department of Biotechnology & Bioinformatics, Yogi Vemana University, Kadapa-516 005, Andhra Pradesh, India

***Corresponding author**

Abstract

The present study focuses on biosorption efficiency of carbonized seed of *Sapindus emarginatus* for the removal of safranine and methylene blue from aqueous solution. The effect of parameters such as initial concentration, contact time, adsorbent dosage, pH and temperature were studied. The optimum adsorption capacity of 4.48 mg/g and 6.9 mg/g was achieved for the safranine and methylene blue dye respectively, from isotherm studies. It revealed that the equilibrium data are fitted well with the Dubinin-Radushkevich and Freundlich isotherm while the adsorption kinetic data showed good agreement with the pseudo – second order kinetic model. Thermodynamic parameters such as ΔG° , ΔH° , and ΔS° for the adsorption process were also determined.

Key Words: Safranine, Methylene Blue, *Sapindus emarginatus*, Adsorption Isotherm and Kinetics.

INTRODUCTION

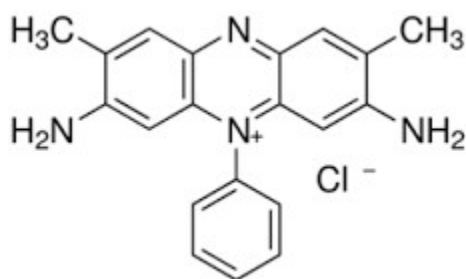
Waste water discharged from anthropogenic activities contains several contaminants. Textile, plastic, cosmetics, printing, paper personal care products, dyestuff, pesticides, heavy metals and pharmaceutical industries may pose negative consequences to the environment due to hazardous chemicals and dyes in effluents. Dyes play a major role in reduction of water quality as they are produced at a significant amount per year and used widely in textile, food coloring, paper and tannery industries (6). Dyes contaminate both surface and ground water causing serious environmental threats. Therefore safranine and methylene blue should be removed from industrial effluents. This is toxic and it is carcinogenic and mutagenic in nature. Because of

change in color, photosynthesis process gets disturbed consequently, aquatic life gets disturbed completely. Safranine and methylene blue are cationic dyes. These are the major pollutants in wastewater streams released from the industrial activities which effects directly on human health such as lungs, throat, kidney, liver, skin infection and irritation of eyes. Removal of dyes from water bodies is a major challenge. Numerous expensive methods such as flocculation, ozonation, oxidation, precipitation, electrolysis, ion exchange, membrane separation, sepiolite, montmorillonite and vermiculite, were used to remove dyes from aqueous solution. Out of all, the activated carbon is the most commonly used for the adsorption process in industrial waste water treatment due to its abundant pore structure and large specific surface area (9). Biosorption technique has been found to be efficient, viable and economically feasible. Some of the adsorbents that have been used to remove safranin dye includes *Mangifera indica* (1), *Moringa oleifera* (2), Pineapple peels (18), fly ash and orange peel (12) and respectively. Various adsorbents for methylene blue dye was tea waste (11), almond gum (5), neem leaf (3), *Vallisneria natans* waste (9), In this study carbonized seed of *Sapindus emarginatus* was used to remove Safranin and methylene blue from aqueous solution.

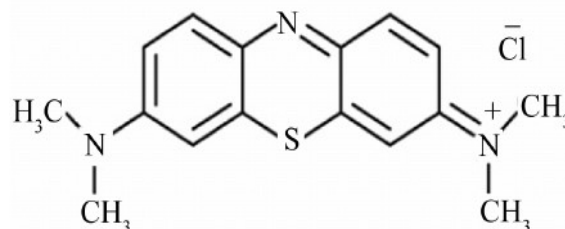
MATERIALS AND METHODS

Adsorbate

Safranin and Methylene blue were purchased from Himedia. The molecular formula of cationic dyes Safranin are $C_{20}H_{19}N_4Cl$ and Methylene blue are $C_{16}H_{18}ClN_3S \cdot 3H_2O$. In order to prepare stock solution, 1.0 g of adsorbate was dissolved in one liter of distilled water. The natural solution pH for safranin (6.0) and methylene Blue (7.0) was kept constant throughout the experiment.



Safranin



Methylene blue

Preparation of Material

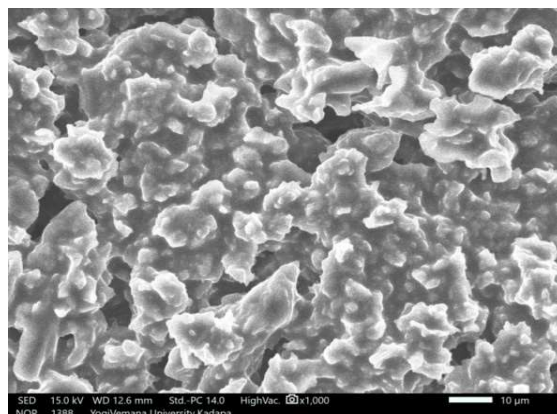
Sapindus emarginatus seeds were obtained from local market in Kadapa, Andhra Pradesh, India. The seeds were washed to remove dust particles and dried in hot air oven at 50°C for 6 hours. It was then ground and sieved with 25 mm sieve. The dried adsorbent was carbonized in hot air oven at 300°C for 90 min and used as an adsorbent.

Batch adsorption

The batch adsorption studies were conducted in 250 ml conical flasks filled with 50 ml of adsorbate at different initial concentration of 10 - 40 mg/L for Safranin and 20 – 50 mg/L for Methylene blue respectively. Dose of 300 mg was added into the solution and agitated at 180 rpm for 80 minutes in a rotary shaker. The solution was separated from the adsorbent by the centrifugation at 4500 rpm for 20 minutes. The absorbance of dye was measured by a double beam UV- Vis Spectrophotometer (Lab India UV 3000+) at 521 nm for safranin and 665 nm for methylene blue.

RESULT AND DISCUSSION

The pH point of zero charge (p^H_{zpc}) was found to be 3.7 for carbonized adsorbent. The Scanning Electron Microscope studies were carried out in order to understand the surface morphology and characterization of adsorbent before and after adsorption of cationic dyes. Fourier Transform Infrared spectroscopy with the digital attenuated total reflectance accessory was used to obtain the FTIR absorbance spectra in the range of 4000-500 cm^{-1} . FTIR was used to characterize the relevant properties (14). Loaded and unloaded FTIR are shown in figs 2(A) & 2(B). Some peaks such as 2925 cm^{-1} and 2115 cm^{-1} showed stretching vibration of C-H in aliphatic, 1636 cm^{-1} stretching vibration is due to skeletal C=C in aromatic ring. Some of the peaks are disappeared and new peaks are appeared. In Safranin the peaks at 1438 cm^{-1} shifted to 1443 cm^{-1} and 605 cm^{-1} shifted to 561 cm^{-1} . In case of methylene blue a new peak appeared at 2364 cm^{-1} . 1438 cm^{-1} shifted to 1385 cm^{-1} and 605 cm^{-1} shifted to 667 cm^{-1} .



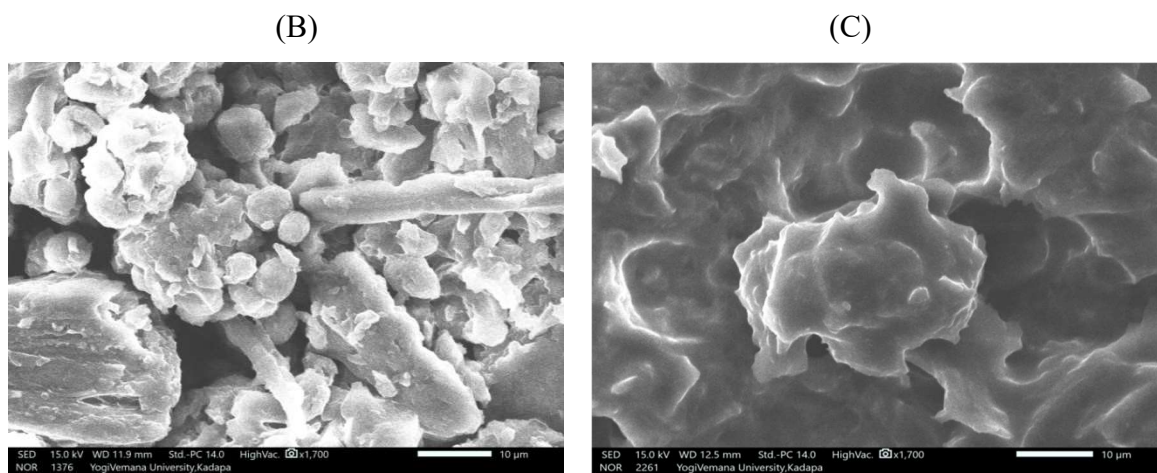
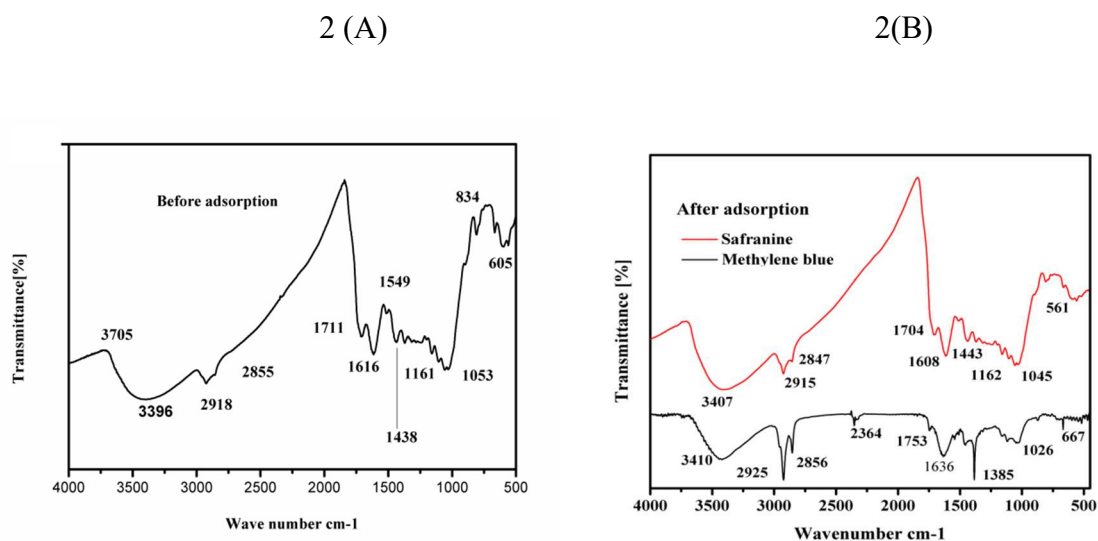


Fig.1 SEM images of carbonized *Sapindus emarginatus* (A) before adsorption (B) safranin loaded adsorbent (C) methylene blue loaded adsorbent.

Fig. 2 FTIR images. (A)Unloaded adsorbent (B) cationic dye loaded adsorbent.



Impact of contact time and initial concentration

The amount (mg/g) of dye adsorbed increased with increase in agitation time and attained equilibrium. The dye removal is directly related to the surface of adsorbent and the contact time. (5) The equilibrium time was 40, 60, min for 10, 20 mg/L and 80 min for 30 and 40 mg/L of initial dye concentration for safranin and 40, 60 min for 20 and 30 mg/L and 80 min for 40 and 50 mg/L of initial dye concentration for methylene blue. The amount of dye removal at equilibrium increased from 1.5 mg/g to 4.1 mg/g for safranin and 3.0 mg/g to 6.0 mg/g for methylene blue with the increase in dye concentration from 10 to 40 mg/L and 20 to 50 mg/L. (Figs. 3 & 4) It was observed that the removal of safranin and methylene blue depends on the concentration. The contact time experiments are useful to investigate the influence of system parameters on the rate and extent of adsorption.

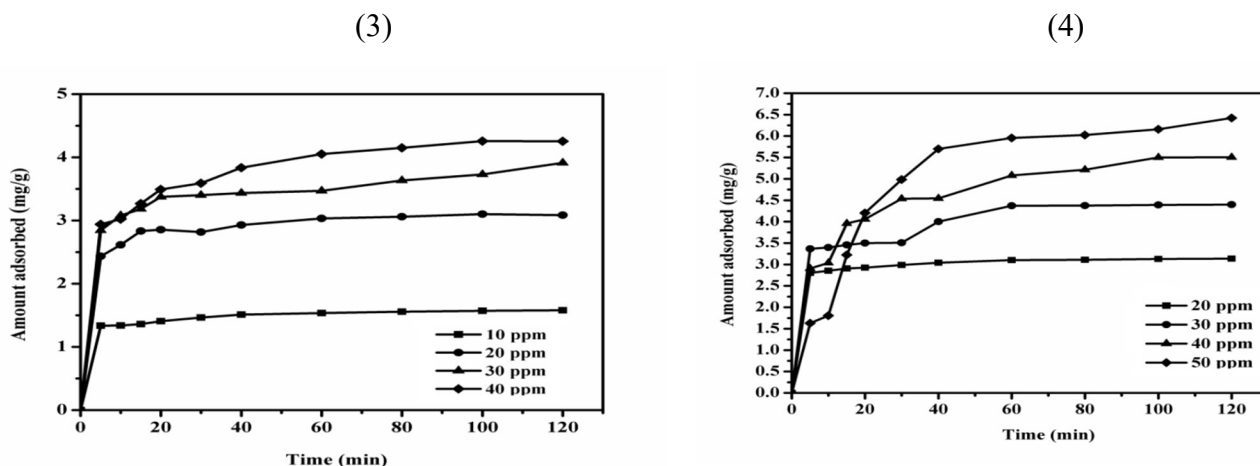


Figure 3, 4 Effect of contact time on Safranin and Methylene blue adsorption.

Adsorption kinetic study

In order to find out the potential rate – controlling steps involved in the process of adsorption, kinetic models i.e., pseudo – first order, pseudo – second order and Elovich models were also established. The pseudo-first order equation was used extensively to describe the adsorption kinetics. Another model for the analysis of adsorption kinetics is pseudo-second-order (Figs 5 & 6; Table 1). In adsorption process pseudo – second order kinetic model data fitted well with the adsorbent. The pseudo second order kinetic model explains that the adsorption process was controlled by chemisorptions involving valency forces through sharing or exchange of electron between the solvent and the adsorbate. (11). The rate coefficient, K_2 , of pseudo – second order was found to decrease with increase in initial dye concentration (Figs. 5 & 6). Similar findings

have been observed in the adsorption of safranine onto pineapple peel (18) and onto African border tree bark (7) respectively.

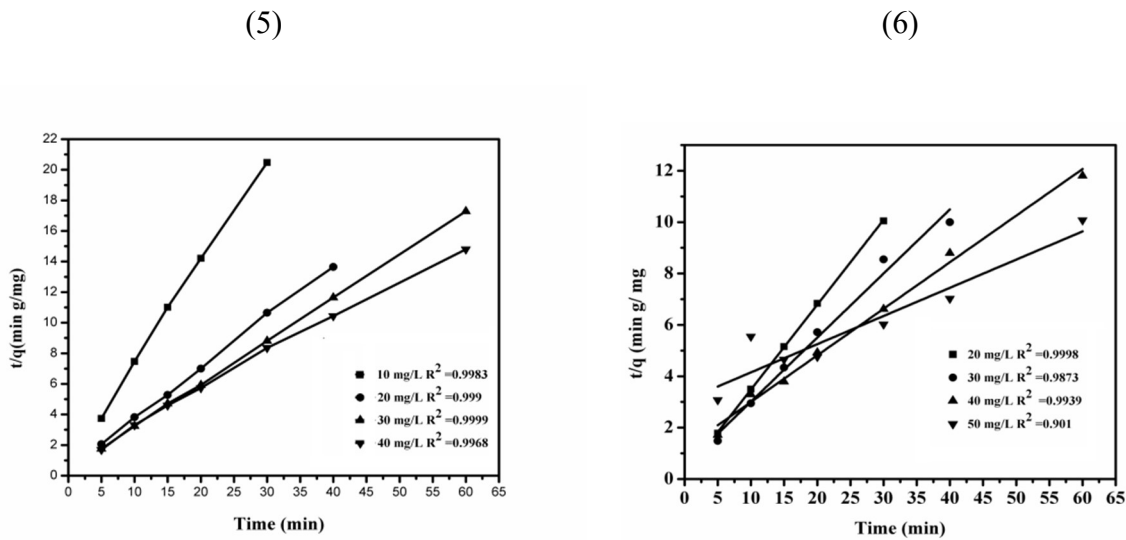


Figure 5, 6 Second-order kinetics plots for the removal of safranine, methylene blue.

Table 1. Comparison of kinetic models of Pseudo first order and second order adsorption rate constants, q_e values of experimental and calculated values for initial concentrations of safranine and methylene blue.

Parameters	First order Kinetic Model				Second order Kinetic Model		
Dye	$q_{eexp}(mg/g)$	$K_1(min^{-1})$	$q_{ecal} (mg/g)$	R^2	$K_2(min^{-1})$	$q_{ecal} (mg/g)$	R^2
Safranine							
10	1.51	0.055	3.54	0.905	0.617	1.5	0.998
20	3.03	0.041	1.75	0.775	0.100	2.99	0.999
30	3.64	0.025	1.50	0.816	0.073	3.56	0.999
40	4.15	0.043	0.61	0.96	0.052	4.26	0.996
Methylene blue							
20	3.041	-0.057	0.327	0.984	0.603	3.027	0.999

30	4.373	-0.024	1.292	0.705	0.121	4.008	0.987
40	5.212	-0.049	3.155	0.95	0.276	5.515	0.993
50	6.026	-0.078	8.446	0.987	3.941	9.115	0.901

Adsorption Isotherms

The adsorption isotherm expresses the specific relation between the concentration of adsorbate and its degree of accumulation onto adsorbent surface at constant temperature. Langmuir, Freundlich and Dubinin- Radushkevich (DR) isotherms were frequently used to interpret the adsorption data. Langmuir isotherms were obtained by agitating the adsorbent of fixed dose and the safranin of different concentrations for a contact time greater than the equilibrium time. The constant Q_0 signifies the monolayer adsorption. It shows good agreement of experimental data with Langmuir plots for safranin (Fig 7). All experiments were done at room temperature at 32°C, the adsorption capacity, Q_0 , for safranin was found to be 4.48 mg/g. This is in accordance with the safranin – o adsorption onto spent coffee ground 3.76 mg/g (8) and fly ash 1.76 mg/g (4). Methylene blue data showed good fit to the Freundlich isotherm and it can be used to elucidate a heterogeneous system (Fig 8). The Freundlich constants k_F and n representing the adsorption capacity and intensity indicators respectively (3, 5). Different methylene blue adsorbents capacity reported in literature is Poly vinyl alcohol 2.75mg/g (16) and Brazil nut shells 7.81mg/g (15). The Langmuir and Freundlich model coefficients are listed in table 2&3. The value of E , the mean free energy of safranin and methylene blue on carbonized *Sapindus emarginatus* are reported in Table 4 which is in the range 16 – 40 kJ/mol corresponding to chemical interaction.

Table 2. Langmuir model coefficients

C _o (mg/L)	Q _o (mg/g)	b (L/mg)	R _L
Safranine			
10	4.48	0.70	0.124
20			0.066
30			0.045
40			0.034
Methylene blue			

20			0.104
30	6.915	0.429	0.072
40			0.055
50			0.044

Table 3. Freundlich model coefficients.

Dye	k_f (mg ^{1-1/n} L ^{1/n} g ⁻¹)	n	R ²
Safranine	1.936	3.300	0.781
Methylene blue	2.660	3.149	0.969

Table 4. Dubinin – Radushkevich model coefficients.

Dye	Q _m (mg/g)	E (kg/mol)	R ²
Safranine	93.23	28.86	0.784
Methylene blue	27.024	50	0.954

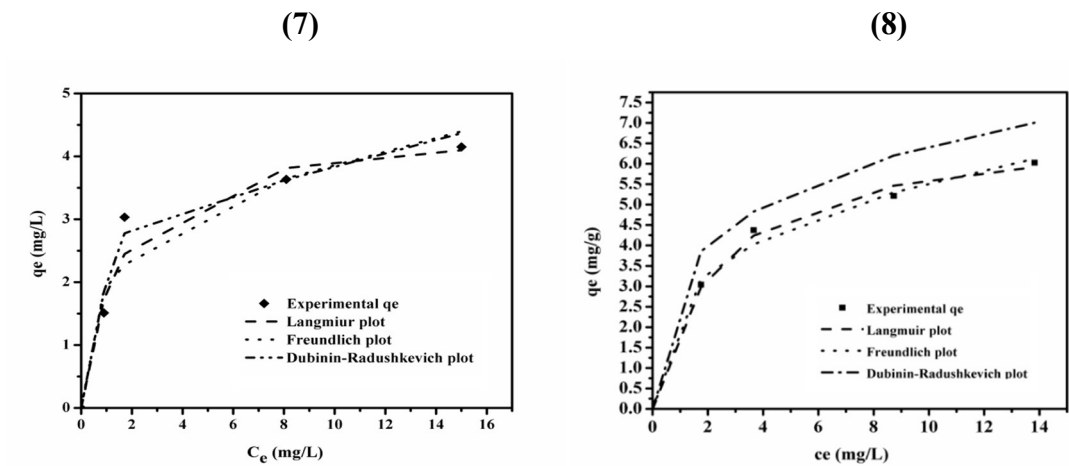


Fig. 7, 8 Comparison of adsorption isotherms of safranine and methylene blue dye.

Impact of pH and desorption studies

Concentrations of ions are considered as one of the vital parameters that influence the adsorption of dyes. The maximum cationic dye removal was recorded at pH 6.0 and 7.0 for safranin and methylene blue dye respectively. The zero point charge (pH_{zpc}) of carbonized *Sapindus emarginatus* was found to be 3.7. The adsorption sites on carbonized *Sapindus emarginatus* less than pH 3.7 is positive in nature and greater than 3.7 is negative. With the increase of solution pH, negative surface charge molecules of carbonized *Sapindus emarginatus* get attracted with positive charge of safranin and methylene dye molecules under electrostatic force of attraction and it favors greater adsorption of safranin and methylene blue dyes. When the solution presents a pH value lower than the pH at the pH zpc, materials present a positive surface charge value. However, when the pH value of the solution is higher than its pH zpc, adsorbents have a negative surface charge value (13). Furthermore, the adsorption behavior of such adsorbent at various pH may be due to surface charges and availability of active sites. Similar observation was noticed earlier (18). Desorption studies show that regeneration of spent adsorbent and recovery of the safranin and methylene blue dyes would make the treatment process economical. Further, desorption studies help to elucidate the mechanism of adsorption. Incomplete desorption of safranin and methylene blue at different concentration confirmed that the chemisorptions was the major mode of adsorption.

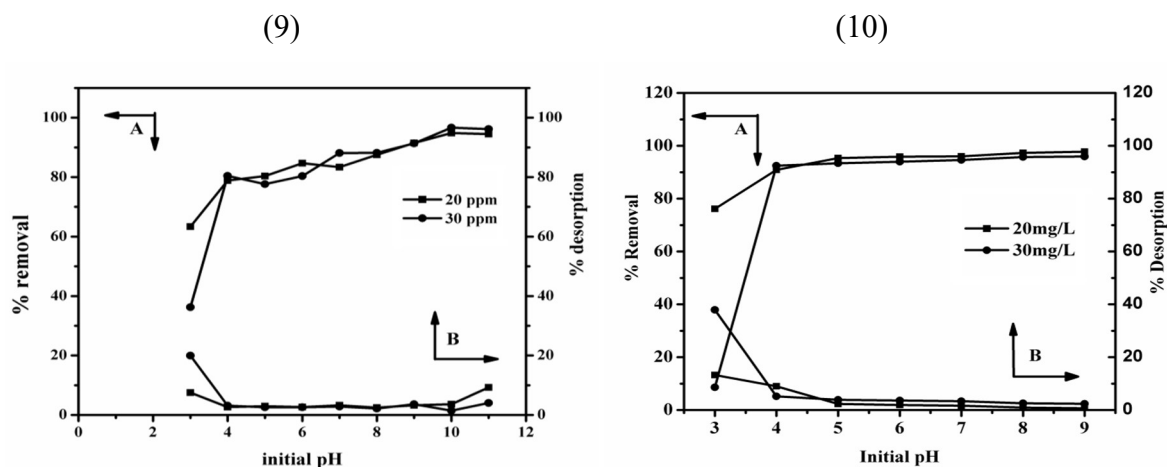


Fig.9,10 Effect of pH on adsorption and desorption of safranin and methylene blue

Studies of thermodynamics of adsorption

Thermodynamic parameters for the adsorption of safranin and methylene blue onto carbonized seed of *Sapindus emarginatus* was calculated by the following equation and were reported in Table 5.

$$\Delta G^{\circ} = -RT \ln K_c \quad (1)$$

Where R is gas constant (8.314 J/mol/K), b is equilibrium constant (L/mol) and T is the temperature in (K):

$$\log K_c = (\Delta S^{\circ} / 2.303 \times R) - (\Delta H^{\circ} / 2.303 \times RT) \quad (2)$$

The values of ΔH° and ΔS° were determined from the slope and intercept of van't Hoff plots $\log K_c$ vs $1/T$. Negative values of ΔH° suggest the exothermic nature of adsorption. The negative values of $\Delta G^{\circ} (< -40 \text{ kJ mol}^{-1})$ indicate spontaneous nature of adsorption and beneficial activity under the experimental conditions (410, 17). It involved in strong chemical reaction. The positive value of ΔS° with *Sapindus emarginatus* indicate increased randomness at the solid solution interface.

Table 5. Co-efficient of adsorption thermodynamics.

Dye	$\Delta H^{\circ}(\text{KJ/mol})$	$\Delta S^{\circ}(\text{J/mol/K})$	R^2
Safranin	-74.95	252.148	0.8158
Methylene blue	-43.67	149.1	0.7744

CONCLUSION

The study informs that *Sapindus emarginatus* acts as a good adsorbent for the removal of safranin and methylene blue from aqueous solutions. The adsorption kinetics data followed pseudo-second order kinetic model with high correlation coefficient almost reaching a unit value for safranin and methylene blue. The removal as a result increases with increase in adsorbent dosage. The optimum operational pH was evaluated as 6.0 for safranin and 7.0 for methylene blue respectively. The use of this low - cost adsorbent can be successfully adopted in treating large amount of dye wastewaters.

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